

Department: **Data Science and Artificial Intelligence**
Course Name: **Data Science Tools and Software**
Course Code: **03203N**
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Level: **Two**



Sheet 1
(Big Data & R Programming Review)

[1] Choose the correct answer :

(1)refers to the number of types of data.

- | | |
|------------|--------------|
| a.Variety | b.Volume |
| c.Velocity | d.Complexity |

(2)..... a data that is not organized or does not have a pre-defined data model , thus it is not a good fit for a relational database.

- | | |
|---------------------|------------------------|
| a.Structured data | b.Semi-structured data |
| c.Unstructured data | d.All of the above |

(3) are any facts, numbers, or text that can be processed by a computer.

- | | |
|-------------|---------------------|
| a.Data | b.Information |
| c.Knowledge | d.None of the above |

(4) Data scientists make big data instructed format even if it is unstructured.

- | | |
|--------|---------|
| a.True | b.False |
|--------|---------|

(5) Big Data is just about lots of data

a.True

b.False

(6) Complexity is one of characteristics of big data which is defined by data speed

a.True

b.False

(7)systems were designed to address smaller volumes of structured data , or a predictable , consistent data structure.

a.Traditional database

b.big data

b.Distributed database

d.None of the above

(8) The negative effects of noisy data can be smoothed using.....

a.Velocity data

c.Veracity data

b.Volume data

d.variety data

(9)Of Big data means that Data is being generated fast and need to be processed fast.

a.Velocity data

c.Veracity data

b.Volume data

d.variety data

[2] Solve the following questions using R :

(1) Explain the output(s) of the following R commands:

a- `> 4*`
`+ 5`

b- `x <- c(1, 2, 3); y <- c(10, 20); z <- c(x, y, 5); u <- c(y^2, x)`

c- `rnorm(10) -> k;`

(2) Generate a sequence of integer numbers from 0 to 10 with step 1

(3) Generate a sequence of integer numbers from 10 to 100 with step 5

(4) Generate a sequence of integer numbers from 20 to 10 with step 2

[3] Describe the result(s) of the following R commands:

- `x<- seq(1,20,2) ; y<- x<= 11;`
- `A<- c(rep(1,3), rep(2,4)) ; B<- A != 1 ; mode(B);`
- `Res= rep(2:4,c(3, 0,2)) ; Res; res= Res >= 4 ; !res`
- `matrix(c(1,2,3,10,20,30,100,200,300),nrow=3) -> A
det(A); B<- t(A);`

e. `A1<- c(2,-2,22); diag(A1); dim(A1); dim(diag(A1));`

f. `A <- matrix(c (1 ,3 ,3 ,9 ,6 ,5) ,2 ,3)
B <- matrix(c (9 ,8 ,8 ,2 ,9 ,0) ,2 ,3)
rbind(A,B); cbind(B,A)
diag(A) -> d1; sd<- diag(A)+2; mb<- B*3;`

g. `y<- c(T,F,F,T); x <- (F,F,T,T)
z1 <- x&y ; z2 <- x|y`

h. `c(1:5, pi, "text")`

i. `letters[10:15] -> f`

[4] Compute the intersection (logical and numeric) of {1, 2, . . . , 10} and {5, 6, . . . , 14}

*Good Luck
M. Samy*